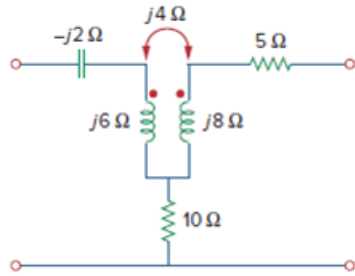


Problem

Find the z parameters of the two-port in Fig. 19.72.

Figure 19.72



Step-by-step solution

Step 1 of 5

Consider the following circuit diagram:

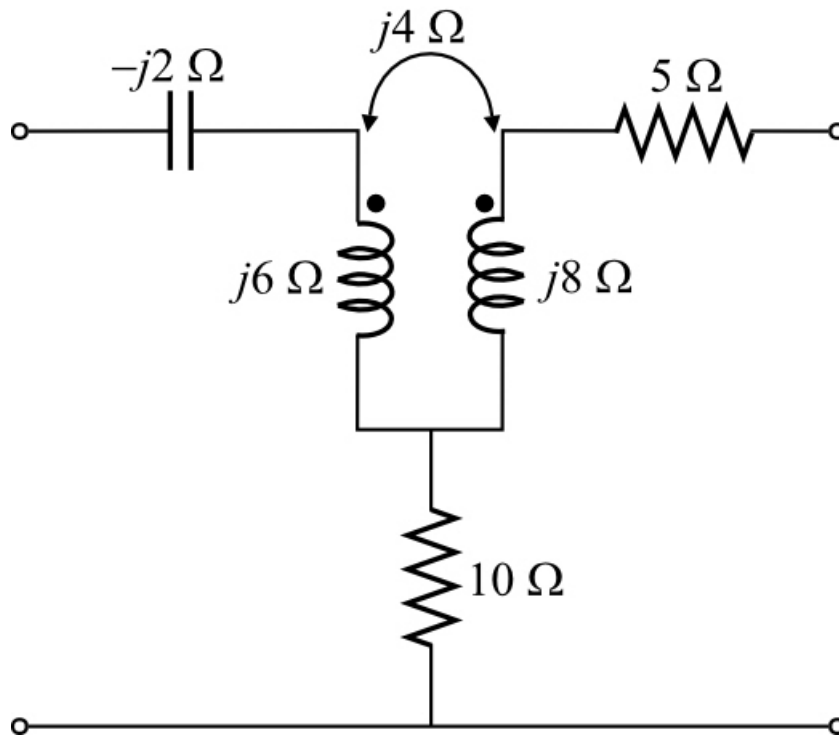


Figure 1

Step 2 of 5

Write the z parameters.

$$z_{11} = \left. \frac{V_1}{I_1} \right|_{I_2=0} \dots\dots (1)$$

$$z_{12} = \left. \frac{V_1}{I_2} \right|_{I_1=0} \dots\dots (2)$$

$$z_{21} = \left. \frac{V_2}{I_1} \right|_{I_2=0} \dots\dots (3)$$

$$z_{22} = \left. \frac{V_2}{I_2} \right|_{I_1=0} \dots\dots (4)$$

Step 3 of 5

To calculate parameters z_{11} and z_{21} , connect a test voltage source to input port and open circuit the output port. Draw the modified circuit diagram.

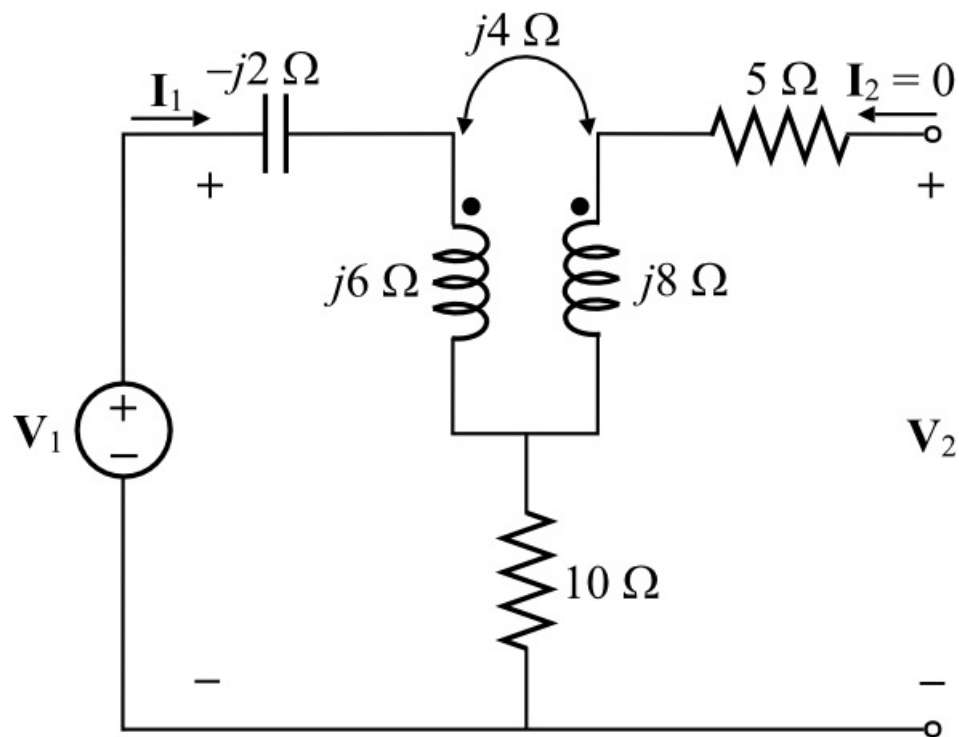


Figure 1

Step 4 of 5

Apply Kirchhoff's voltage law to left side loop.

$$V_1 = (-j2 + j6 + 10)I_1$$

$$\frac{V_1}{I_1} = 10 + j4$$

$$z_{11} = 10 + j4$$

Apply Kirchhoff's voltage law to right side loop.

$$V_2 = (j4 + 10)I_1$$

$$\frac{V_2}{I_1} = j4 + 10$$

$$z_{21} = j4 + 10$$

To calculate parameters z_{12} and z_{22} , connect a test voltage source to output port and open circuit the input port. Draw the modified circuit diagram.

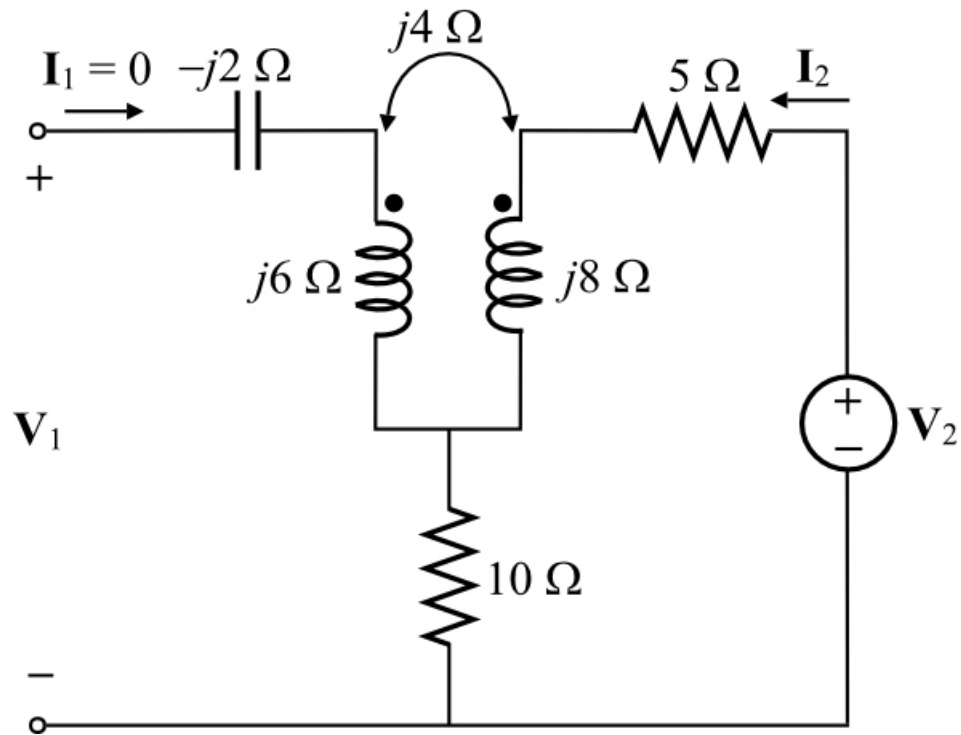


Figure 3

Apply Kirchhoff's voltage law to left side loop.

$$V_1 = (j4 + 10)I_2$$

$$\frac{V_1}{I_2} = 10 + j4$$

$$z_{12} = 10 + j4$$

Apply Kirchhoff's voltage law to right side loop.

$$V_2 = (5 + j8 + 10)I_1$$

$$\frac{V_2}{I_1} = 15 + j8$$

$$z_{22} = 15 + j8$$

Therefore, the z parameters, $\begin{bmatrix} z_{11} & z_{12} \\ z_{21} & z_{22} \end{bmatrix}$ are $\begin{bmatrix} 10 + j4 & 10 + j4 \\ 10 + j4 & 15 + j8 \end{bmatrix}$.